CCE Unit-3
Linked List.

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* Title : Bus Route Management Using circular Linked List.

1. Problem statement :
In a circular Bus Route, the bus travels through a fixed sequence of a starting stop & Returns to the starting stop after reaching the last stop. The Bus stop needs to be represented efficiently such that the last stop is connected Back to first stop. Implement a circular clinked list to represent & Display the Bus Route.

2. Solution :
A circular linked list is used to represent a bus stop and pointer to the next stop. The next stop of the last node points Back to the first node, forming a circular route.

3) Objectives :

- To Represent a bus stop using a circular linked list.
- To display all Bus stops by performing circular traversal.
- To understand the use of nodes & pointers in a circular linked list.
- To demonstrate how the last Bus stop connects Back to first stop.
- To understand the practical application of a circular linked list in Bus Route management

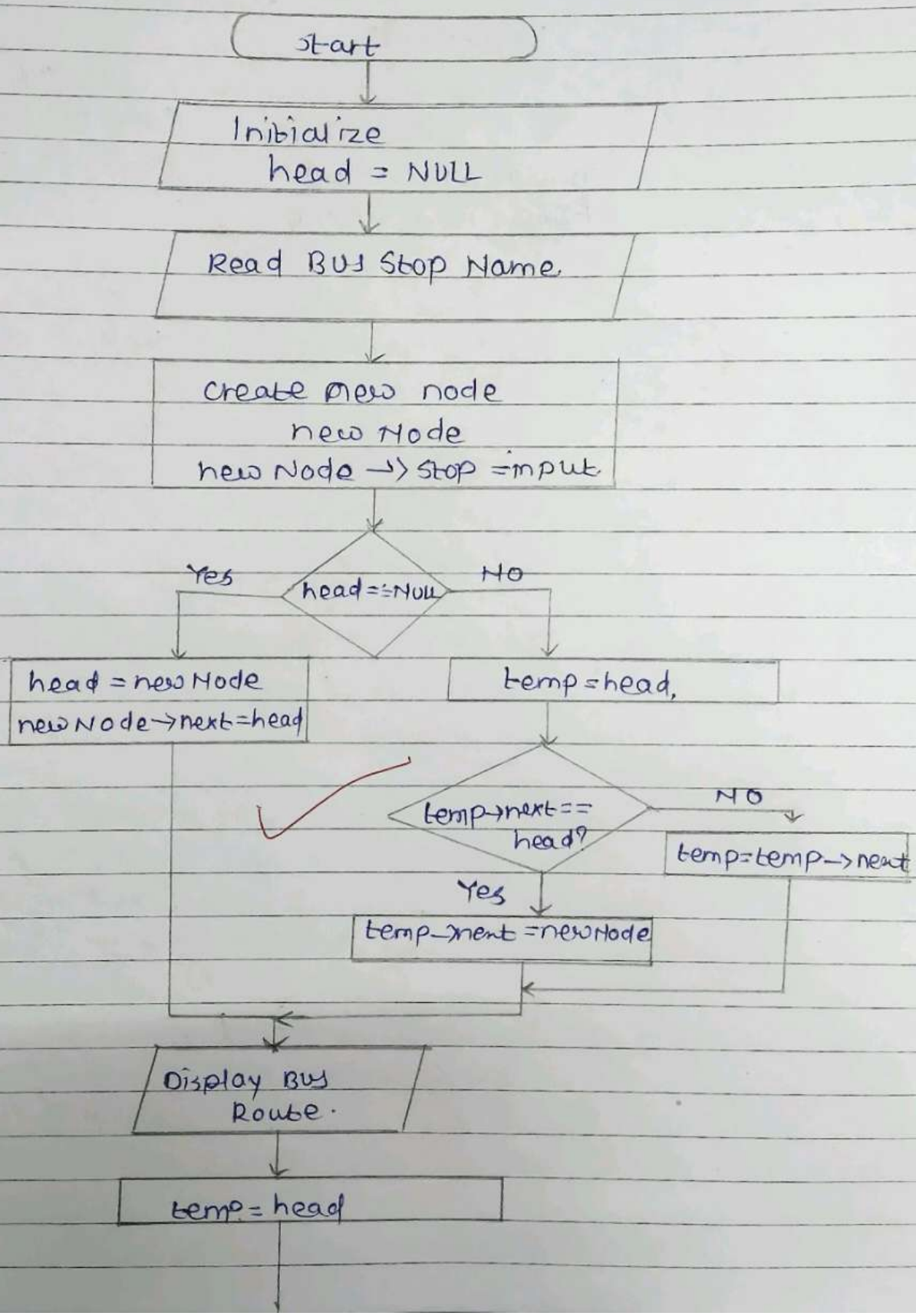
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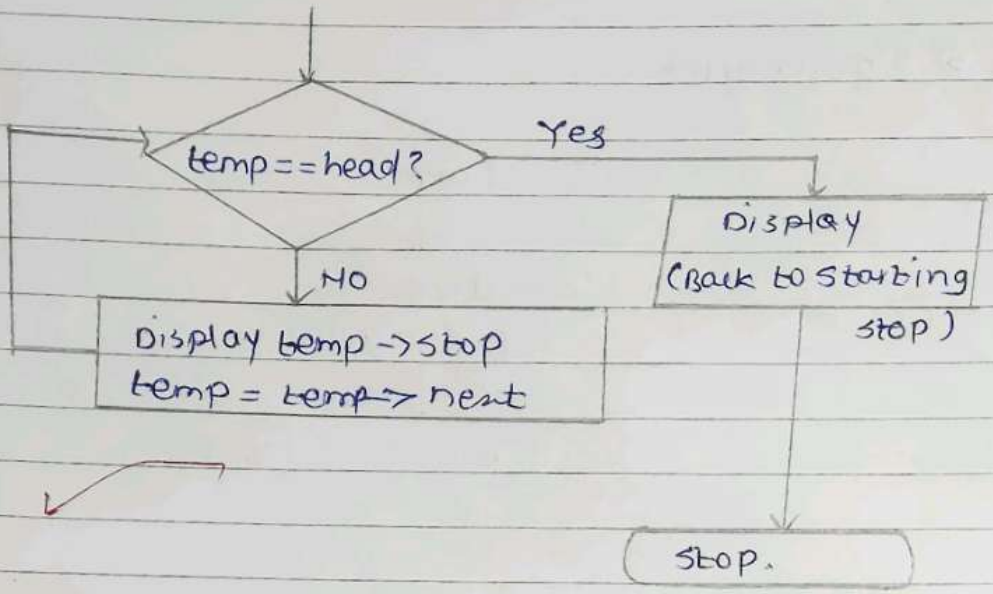
4) Algorithm :-

Step 1 : start.
Step 2 : Define a node containing:
Bus stop name.
pointer next.
Step 3 : Initialize head = NULL.
Step 4 : Read the Bus stop name to be inserted.
Step 5 : create a new node and store the Bus stop name in it.
Step 6 : if head = NULL :
Set head = new Node
set newNode -> next = head.
Step 7 : otherwise :
Traverse the list until temp -> next == head.
set temp -> next = newNode.
set newNode -> next = head.
Step 8 : to display the route, start from head.
Step 9 : Display each Bus stop and move to the next node until temp == head.
Step 10 : Display that the last stop connects Back to the First stop.
Step 11 : stop.

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5) Flowchart :-




temp==head?
Yes
No
Display temp->stop
temp = temp->next
Display
(Back to starting
stop)
Stop.